NEEL TILAK

Software Engineer | Backend & Cloud Infrastructure | 0-to-1 Systems Builder

Mumbai, India | +91 7303839197 | neeltilak.1998@gmail.com | linkedin.com/in/neel-tilak-74734b231

PROFESSIONAL SUMMARY

Software Engineer with 3+ years of experience specializing in backend architecture, distributed systems, and private cloud platforms. Proven 0-to-1 systems builder who has architected and delivered enterprise production systems from scratch across Cloud IaaS, Luxury E-Commerce, Fintech, and CRMs. Deep, active hands-on expertise in C#, ASP.NET Core 8, Entity Framework Core (EF Core), RabbitMQ, Proxmox VE virtualization, and high-availability networking (HAProxy, SDN, OPNsense). Holds a B.E. in Computer Science with a 9.4 CGPA (First Class with Distinction).

TECHNICAL SKILLS

- **Core Backend (Daily):** C#, ASP.NET Core 8, Entity Framework Core (EF Core), Web API, RabbitMQ, BackgroundService Workers, RESTful APIs
- **Cloud & Virtualization:** Proxmox VE (PVE 8.x), QEMU Virtual Machines (KVM), LXC Containers, Docker, Cloud-Init, Linux (Ubuntu)
- **Networking & High Availability:** HAProxy (L4 Load Balancing), OPNsense (1:1 NAT, DNAT, Firewalls), SDN (VXLAN/EVPN), NGINX Ingress, TCP/IP, DNS
- **Databases & Observability:** Microsoft SQL Server, PostgreSQL, MySQL, SQLite, Prometheus, Elasticsearch, Serilog
- **Client Delivery & Mobile:** Flutter, WordPress & WooCommerce Custom Plugin Development, API Testing, Postman, Git, Azure AD OAuth, JWT, 2FA
- **Familiar / Secondary Stack:** Node.js, Express.js, Python (FastAPI), Go, React.js, Angular, ADO.NET (Stored Procedures)

PROFESSIONAL EXPERIENCE

Hirdhav Software Pvt Ltd | Software Developer / Cloud Infrastructure Engineer — **March 2024 – Present | Mumbai, India**

Core Platform: Indryom (Private Cloud Infrastructure Platform — IaaS)

- **Virtualization & Compute:** Architected core backend APIs and domain models using ASP.NET Core 8, C#, and Entity Framework Core (EF Core), automating lifecycle provisioning, dynamic disk sizing, and teardown of KVM VMs and LXC Containers via Proxmox VE REST APIs in under 30s.
- **ORM & State Persistence:** Designed high-performance data access layers and migrations with Entity Framework Core (EF Core) and SQL Server, enforcing strict transactional consistency across cloud tenant resources.
- **High Availability & L4 LB:** Architected Layer 4 HAProxy load balancing service with live health checks and round-robin traffic routing, achieving sub-5ms routing latency across scaled backend container replicas.
- **Network Automation & SDN:** Integrated OPNsense appliances for automated 1:1 Public IP NAT, DNAT, and edge firewall rules; designed shared VXLAN overlay bridges and edge NGINX gateway.
- **Storage & State Reconciliation:** Built transactional storage workflows supporting dynamic disk resizing and automated vzdump backups with state reconciliation between SQL Server and physical hypervisors.
- **Workers & Usage Metering:** Built resilient BackgroundService workers and RabbitMQ consumers for real-time compute usage tracking, invoice calculation, and automated billing suspension.
- **Zero-Trust Security & IAM:** Implemented multi-tenant IAM with Azure AD OAuth, TOTP 2FA (AES-256 encrypted), and isolated short-lived identity handoff tokens for cross-portal support.
- **Observability & Metrics:** Centralized application audit logs via Elasticsearch and cluster health telemetry via Prometheus for end-to-end multi-node observability.

Client Product Deliveries (0-to-1 Solo Engineering)

- **SaturnnLGD (Diamond E-Commerce):** Built ASP.NET Core 8 Web APIs, MVC admin portal, and Flutter app; engineered custom WooCommerce & PHP plugins from scratch for automated Blue Dart shipping/waybills and PDF invoicing.
- **KIFS Home Loan Platform:** Developed a Flutter mobile application for digital loan origination, document verification pipelines, and multi-stage approval workflows deployed behind NGINX.

Terado Engineering | Jr Backend Developer — **Dec 2023 – March 2024 | Mumbai, India**

- **Node.js Microservices:** Engineered RESTful backend microservices in Node.js, Express.js, and Sequelize ORM backed by MySQL for Uni5 HRMS (geolocation attendance, employee management) and PTEcolyzer CRM (lead pipelines, automated environmental alerts).
- **API Testing & Optimization:** Executed comprehensive API testing and optimized database query execution plans, reducing API latency across reporting endpoints.

Freelance Software Developer | Freelance Software Engineer — **June 2023 – Nov 2023 | Mumbai, India**

- **Contract Development:** Delivered full-stack web modules and mobile prototypes for private clients using .NET and React; integrated SSO authentication and third-party payment gateways.

EDUCATION

Rajiv Gandhi Institute of Technology (Mumbai University) — B.E. in Computer Science & Engineering (CGPA: 9.4 / 10 — First Class with Distinction) — **Aug 2020 – May 2023**

Pravin Patil College of Diploma Engineering & Technology (MSBTE) — Diploma in Computer Engineering (91.37% — Distinction) — **June 2017 – May 2020**